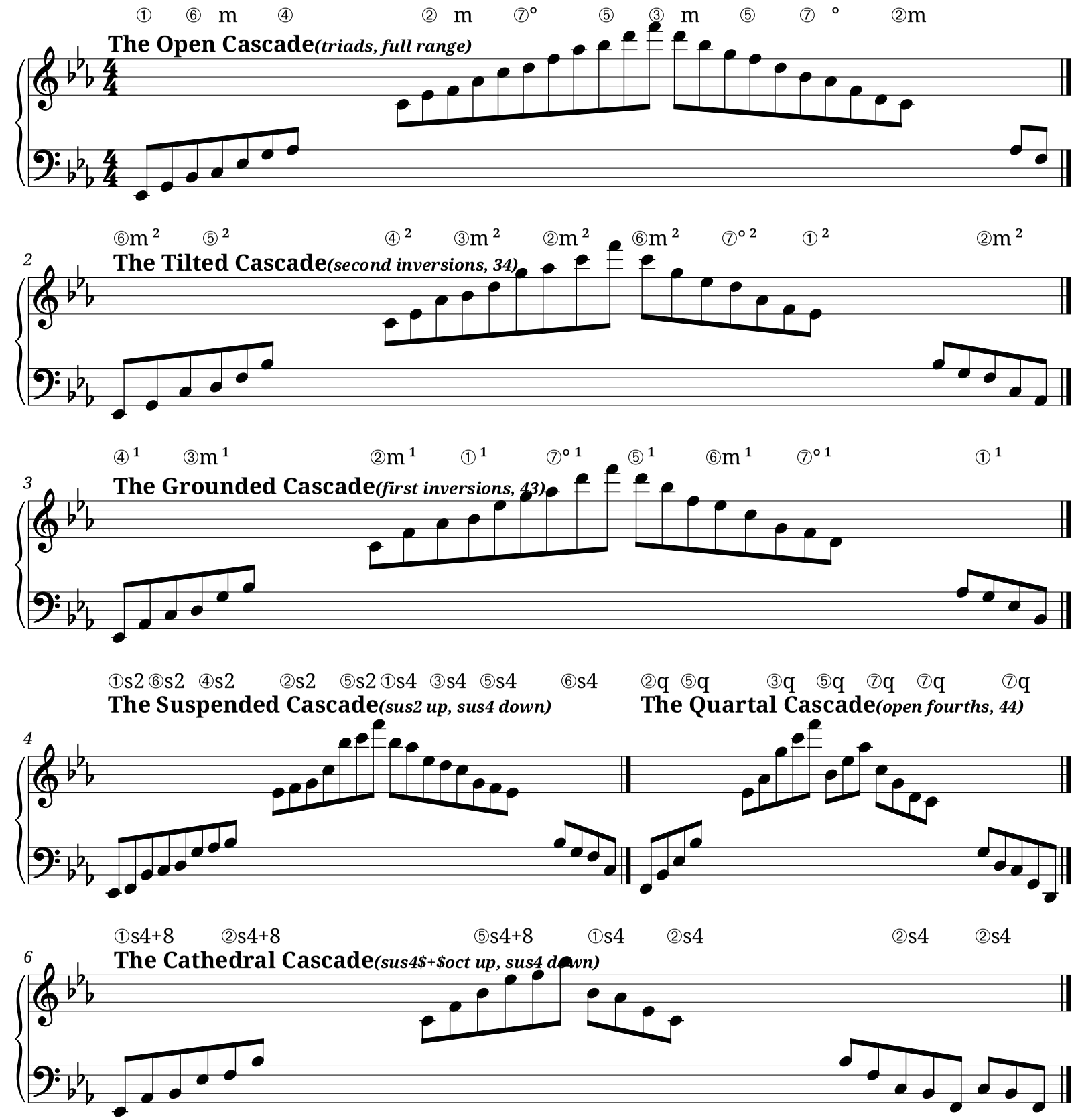
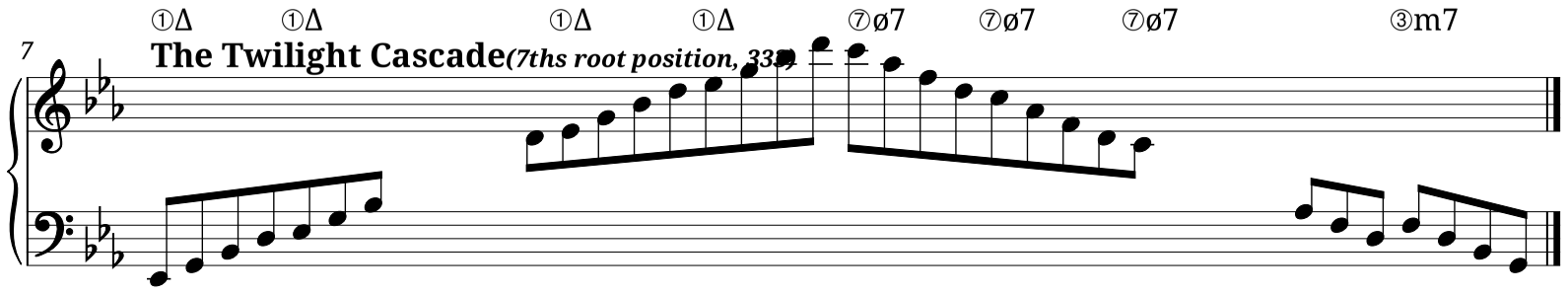
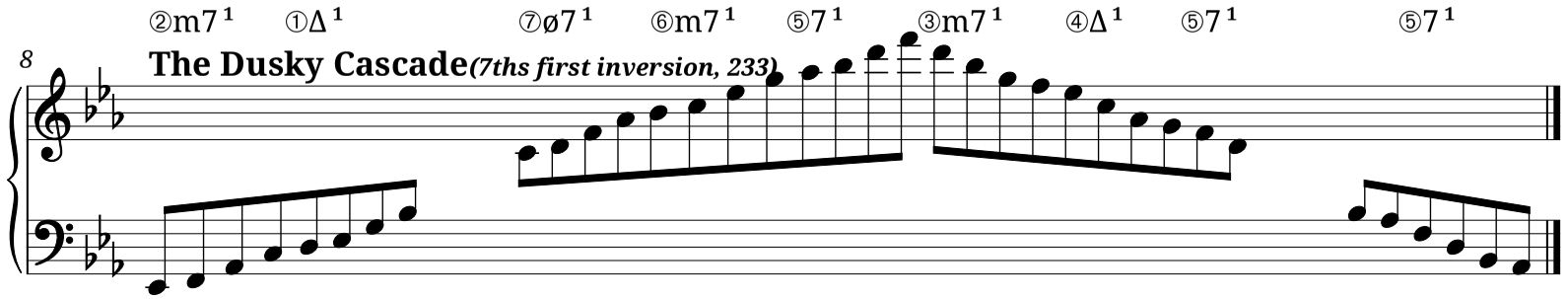
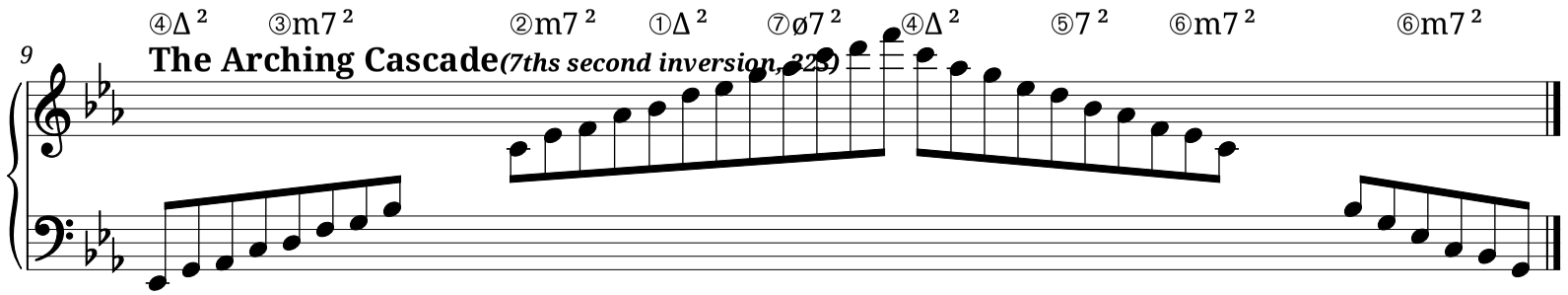
Cascade Reference Sheet

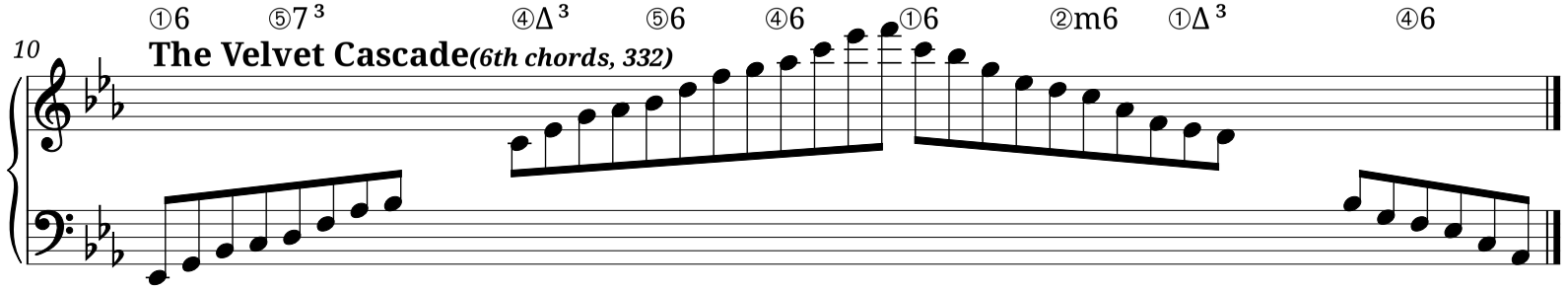


MEI engraved with Verovio


①Δ ①Δ ①Δ ①Δ ⑦ø7 ⑦ø7 ⑦ø7 ③m7
7
The Twilight Cascade(7ths root position, 333)


②m7 1 ①Δ 1 ⑦ø7 1 ⑥m7 1 ⑤7 1 ③m7 1 ④Δ 1 ⑤7 1 ⑤7 1
8
The Dusky Cascade(7ths first inversion, 233)


④Δ 2 ③m7 2 ②m7 2 ①Δ 2 ⑦ø7 2 ④Δ 2 ⑤7 2 ⑥m7 2 ⑥m7 2
9
The Arching Cascade(7ths second inversion, 323)


①6 ⑤7 3 ④Δ 3 ⑤6 ④6 ①6 ②m6 ①Δ 3 ④6
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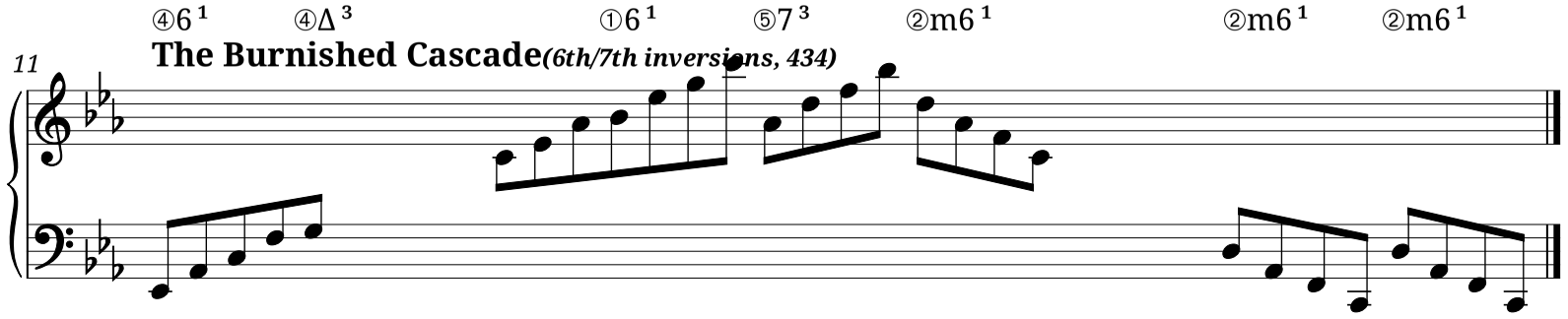
The Velvet Cascade(6th chords, 332)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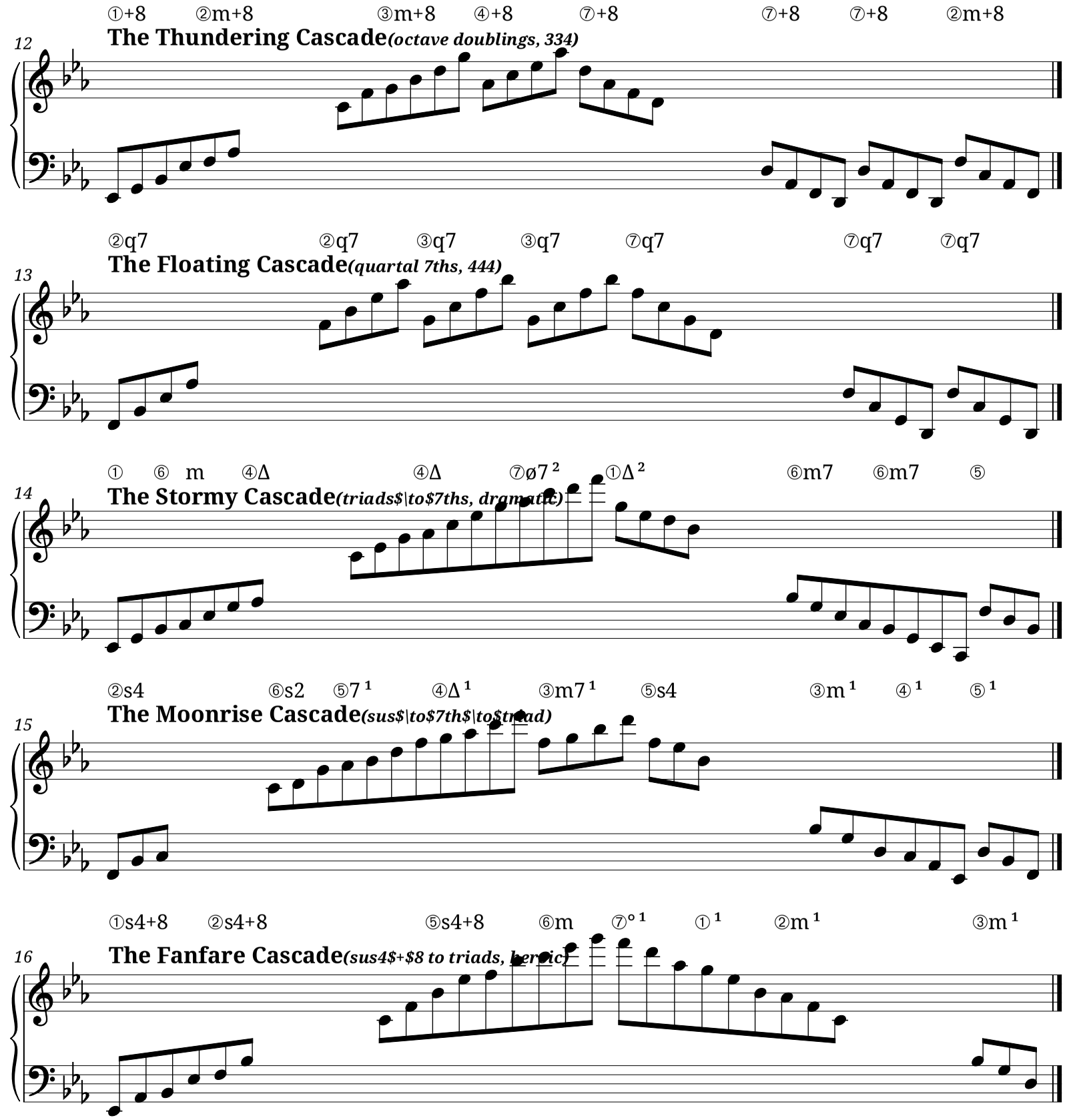
11
The Burnished Cascade(6th/7th inversions, 434)



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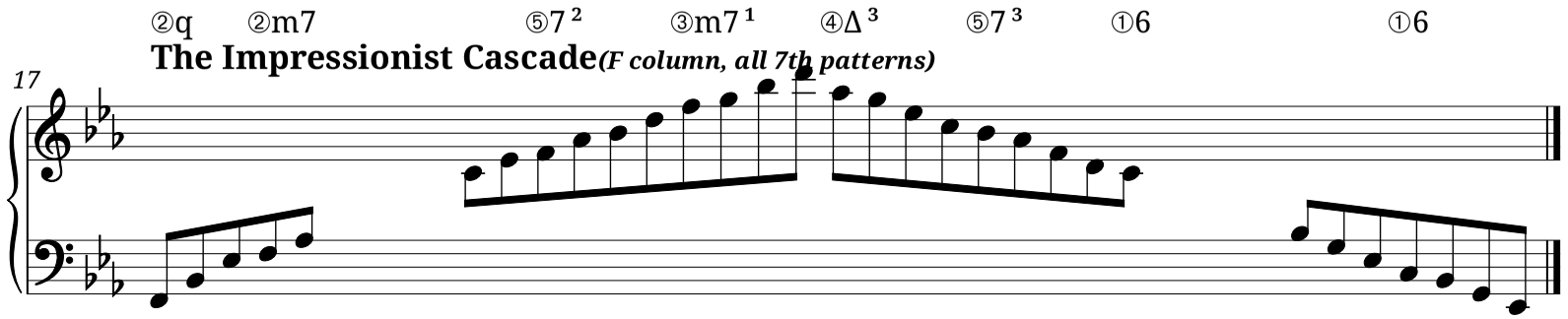

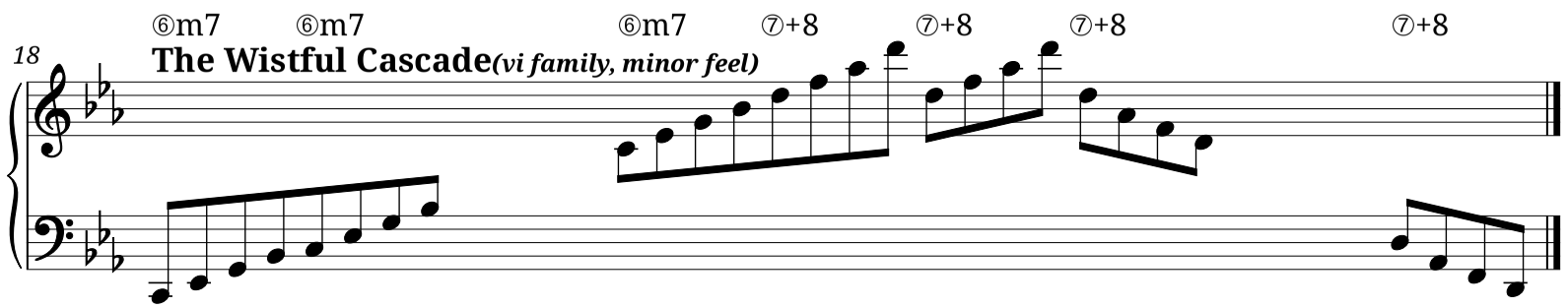

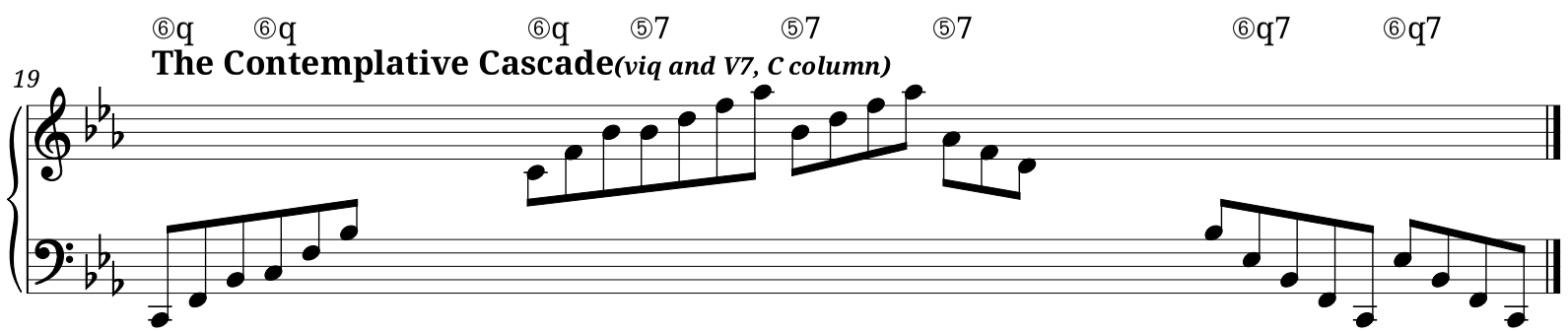

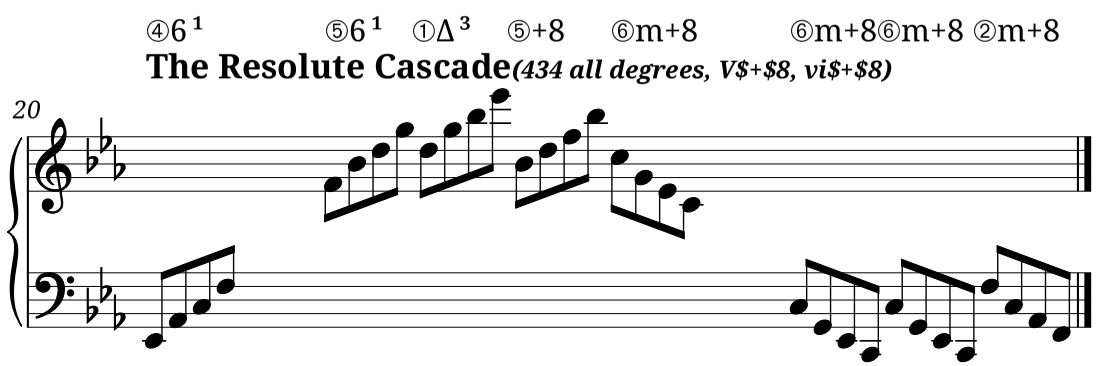

Practice Framework

How to actually improve, every session

The Feedback Loop — every single rep

No rep is wasted if you treat each one as an experiment.

- **Plan**: before playing, set a specific target. Not "play it better" — something like "keep the LH thumb on the string until the RH note starts."
- **Play**: execute the target honestly, at whatever tempo lets you think.
- **Evaluate**: did you hit the target? Be strict. 90% is success; 50% is information.
- **Adjust**: change one variable before the next rep — slow down, shorten the phrase, or pick a narrower target. Never repeat blindly.

Liszt: "*When practicing, think ten times and play once.*"

The 90/10 Principle

To nail a skill at 100%, stop grinding it head-on. Push a **harder related skill** to 90% instead — the easier one snaps into place automatically.

Examples for harp:

- Want clean arpeggios at tempo? Practice them with finger-numbers said out loud at 70% tempo. When you stop speaking, tempo goes up.
- Want a smooth LH/RH hand-off? Do the cascade in reverse (descending then ascending). When you go forward again it feels easy.
- Want even dynamics? Practice crescendos and decrescendos over the same notes. Then "flat" feels natural.

You're not doing the target skill — you're doing *something harder*. When you remove the extra layer, the target skill runs on less effort.

Five Axes of Difficulty

When "harder" is vague, pick one of five axes. Raise one, lower another, cycle.

Length	one measure → two → phrase → whole piece. Endurance *is* a skill.
Complexity	one hand → two hands; melody alone → melody + chords; add a cascade across a bar line.
Depth	not just notes: phrasing, dynamics, articulation, tone. The "musicality" layer.
Speed	only after the first four feel solid. Speed is the easiest axis and the most overused.
Mental endurance	how long you stay focused *inside* the playing. Sustained attention, not just physical stamina.

3-4-2-1 Session Order

Your brain's focus falls during any session. Front-load the hardest work.

Rank every chunk of material into four levels:

- **Level 4** — "impossible right now." Can't play it cleanly even once.
- **Level 3** — "hard." You have to be at your A-game.
- **Level 2** — "almost there." Mostly clean, occasional slip.
- **Level 1** — "automatic." Plays itself.

Session order: 3 → 4 → 2 → 1. Start at level 3 as a warm-up that demands real focus (skip level-1 warm-ups — they set your brain to cruise). Move to level 4 while you're sharpest. Drop to level 2 as attention fades. End with level 1 for review, while your brain is already in "just play" mode.

This works at any skill level — your level 4 as a beginner and a professional's level 4 are different pieces, but the structure is identical.

Chopin told his students two hours a day was enough. They learned his etudes. The difference was never quantity.